Electrical Power and Energy Systems 154 (2023) 109457

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Reed switch protection of double-circuit lines without current and voltage transformers

M.Ya. Kletsel, B.E. Mashrapov *, R.M. Mashrapova

Toraighyrov University, Lomov str., 64, Pavlodar 140008, Kazakhstan

ARTICLE INFO

Keywords:
Protection of double-circuit lines
Sequence of actuation of the reed switches
Selection of operation parameters
Algorithm of operation
Determine the fastening point
Structure for fastening

ABSTRACT

Protection of 6–35 kV double-circuit lines from the supply side is suggested. It allows saving copper, steel, and high-voltage insulation in amounts unprecedented for relay protections. The protection is based on the control of sequence of actuation of reed switches fastened near the corresponding phases at a safe distance and of the time Δt between their actuation. The effect of all possible errors on Δt is studied and the number of reed switches required to be mounted near the phases of the lines is determined. The peculiarities of selection of Δt and inductions for the protection operation, points of reed switch mounting, sensitivity estimates, and the fields of use of the protection are analyzed. An example of the selection is considered. The protection operation algorithm is formulated and written in the form of switching algebra functions with allowance for the protection operability in the case of reed switch failures. The correct operation of the protection is shown in all actual modes of operation of the lines (including the failure of one of reed switches). A construction for fastening reed switches at a safe distance from the line phases is presented.

1. Introduction

Transverse current differential protections with power direction relays have been used to protect double-circuit lines for almost a hundred years. It has several well-known disadvantages: low sensitivity in some cases (due to the need to offset from currents in fault-free phases and full-load currents), the use of voltage circuits, etc. Many improvements have been suggested for the protection, including such [1–3], where several disadvantages are simultaneously eliminated. However, to receive information about currents in the lines, current transformers (CTs) are still used. They contain high-quality steel, copper, and high-voltage insulation; saturate in transient conditions, during which relay protection devices incorrectly operate sometimes leading to major accidents. In view of these disadvantages, CIGRE conferences many times emphasized a need in creating protections [4,5] without CTs, and the problem of constructing them was called one of the fundamentally unsolved problems of the electric power industry. Changing CTs to small-size magnetically sensitive elements (Rogowski coil, Hall sensor, reed switch, etc.) would eliminate the consequences of their saturation and save resources [6]. Works in this direction have been carried out for a long time. There are works where these elements are suggested to be used as a basis for designing the overcurrent (including protection with filters of symmetrical components) [6–18], differential [19–21], centralized [22,23], and remote [24] protections for some electrical installations. However, no protections of double-circuit lines based on these elements have been suggested so far. In this paper, we try to fill this gap. We suggest using reed switches, since, unlike other magnetically sensitive elements, they can simultaneously function as current sensors, a current relays, and analog-to-digital converters and, which is very important in relay protection, transmit signals through control circuits, but not through measuring circuits.

2. Reed switch and some its properties

A reed switch (Fig. 1) is a sealed glass tube (GT) filled with an inert gas. Two or more metal plates (contacts) (PL) are fixed inside; they contact (close) if the magnetic induction $B_{lon} \geq B_{act}$ (B_{act} is the reed switch actuation induction) acts along their longitudinal axis. This induction is calculated as

$$B_{act} = \mu_0 F_{act}/l_c$$

where F_{act} is the magnetomotive force of the reed switch actuation; l_c is the reed switch wrap length; μ_0 is the permeability of vacuum.

Under the action of an alternating magnetic field (MF), the reed switch closes (actuates) and opens the contacts every half-wave with a

* Corresponding author.
E-mail address: bokamashrapov@mail.ru (B.E. Mashrapov).

https://doi.org/10.1016/j.ijepes.2023.109457
Received 27 January 2023; Received in revised form 16 June 2023; Accepted 18 August 2023
Available online 25 August 2023
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Nomenclature

B_{lon} induction acting along the longitudinal axis of a reed switch
B_{act} reed switch actuation induction
F_{act} magnetomotive force of the reed switch actuation
l_c reed switch wrap length
t_{act} action time of reed switch
k B_{lon} to B_{act} ratio
t_r rated time of reed switch actuation
B_A, B_B, B_C inductions of magnetic fields produced by currents in the phases A, B, C
I_A, I_B, I_C currents in the phases A, B, C
α_A, α_B, α_C angles between the longitudinal axis of the reed switch and the induction vectors B_A, B_B, B_C
h_A, h_B, h_C distances between the axes of the phases A, B, C and the center of gravity of the reed switch
Δt time between actuation of reed switches fastened near the same phases of the lines
I_1, I_2 currents in the first (L_1) and second (L_2) lines
B_1, B_2 inductions of magnetic fields (MF) produced by currents I_1, I_2
k_1 ratio of the maximum to minimum short circuit current at the boundary of the cascade action zone
Δt_{th} (Δt_{th1}) time between the instants of closing the contacts of reed switches 1 (3) and 2 (4) required for the protection operation
Δt_{im} time imbalance
ε_1 error appears when a reed switch is mounted at the target point
ε_2 error is due to the difference between the reed switch actuation times
ε_3 error is due to the deviation of B_{act} from the calculated value
t time from the B_{lon} zero-crossing time to the reed switch actuation
B_{th1} induction under which reed switches 1, 2, 9, 10, 17, 18 should actuate
k_2 ratio of B_{lon} to B_{th1}
B_{th2} induction under which reed switches 3, 4, 11, 12, 19, 20 should actuate
Δt_{im1} time imbalance under the action of a MF with $B_{lon} = B_{th2}$ on the reed switches
$B_{load.\max}$ induction amplitude of MF produced by the full-load current in a line
l_{ca} section of the line, where $\Delta t < \Delta t_{th}$ under a SC
B_{f1} induction amplitude of MF acting on a reed switch near the phase of a fault-free line under the current in a fault line after circuit breaker opening from the side of the receiving substation in the case of SC in l_{ca}
$B_{ca.r}$ induction amplitude of MF which acts on a reed switch in the minimal mode of system operation in the
($B_{ca.\min}$) case of a double phase short circuit (SC) at the boundary of cascade action zone before circuit breaker opening from the receiving side (at busbars of the receiving part in the case of double phase SC after the breaker opening)
k_s sensitivity coefficient
B_{f2} induction amplitude of MF produced by the double earth fault current in the phases of the lines after opening the fault line circuit breaker from the receiving side
l line length
$X_{S.\max}$ maximal resistance of the power system
X_L resistance of a line circuit
B_{th3}(B_{th4}) induction under which reed switches 3, 4, 11, 12, 19, 20 (7, 8, 15, 16, 23,24) should actuate
O_1 (O_2) signals for opening breakers Q_1 (Q_2)
G_1–G_{24} signals for actuation of reed switches 1–24
R_1–R_4, R_9–R_{12}, R_{17}–R_{20} signals about reed switches 1–4, 9–12, and 17–20 have actuated the first
Z_1–Z_{24} signals about non opening contacts of reed switches 1–24 for 0.01 s after the actuation
T_1, T_2, T_3, T_4, T_5, T_6 signals about exceed of the time between the actuation of reed switches 1 and 2, 3 and 4, 9 and 10, 11 and 12, 17 and 18, 19 and 20 over a value specified
T_7 (T_8) signal of expiration of 5 ms from the actuation of reed switch 5, 13, or 21 (6, 14, or 22)
P^t memorization operator
F_1, F_2, F_3, F_5, F_7, F_9, F_{10}, F_{17}, F_{18} the fault signals of reed switches 1, 2, 3, 5, 7, 9, 10, 17, 18
T_9 signal about expiration of 5 ms from actuation of reed switch 1
$u_{SC.\max}$ ($u_{SC.\min}$) maximal (minimal) short-circuit voltage of the transformer
r_{spec} specific active resistance
X_{spec} specific reactive resistance
X_S system resistance before the transformer
$I_{load.r}$ rated load current amplitude
$X_{T.\max}$ maximal resistance of the transformer
$I_{SC.L1}$, $I_{SC.L2}$ maximum currents in L_1 (fault) and L_2 in the case of a SC at the boundary l_{ca}
I_{1ca}, I_{2ca} currents in L_1 and L_2 in the case of a double phase SC at the boundary of l_{ca} in the minimal operation mode of the power system
$I_{tr.L1}$ maximum current in L_1 with a short circuit at the boundary l_{ca} (second line is broken)
$I_{ca.\min}$ minimum current in the event of a bus SC in the receiving substation, when one of the lines is broken
$I_{SC.ca}$ current of an earth fault at two points
I_{SC} current of a double phase SC at the boundary l_{ca}
$B_{SC.L1}$, $B_{SC.L2}$ inductions of MFs produced by currents $I_{SC.L1}$ and $I_{SC.L2}$
U_{DC} DC voltage across the induction coil under which the reed switch is actuated
U_{AC} AC voltage across the induction coil proportional to B_{lon}
U rated voltage of the lines
k_3 ratio of B_{th2} to B_{th1}
$I_{SC.\min}$ current in one line circuit in the case of a double phase bus SC in the receiving substation in the minimal system operating mode when both lines are live
$B_{SC.\min}$ induction of a MF produced by the current $I_{SC.\min}$
B_{ca1}, B_{ca2} inductions produced by the currents in the fault and fault-free lines in the maximal system operating mode
$\Delta t_{SC.\min}$ ($\Delta t_{SC.\max}$) time between actuation of reed switches 1 and 2 in the case of a SC at the boundary of l_{ca} under the minimal (maximal) system operation modes
U_i is the voltage applied to the microcontroller input when the contacts of the corresponding reed switch are closed (i = 1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 18, 19, 21, 22)
U_s is the voltage at which the microcontroller considers a signal to arrive at its input
t_1- t_5 are the times counted by timers $T1$–$T5$
t_{et1} is the time required to detect that the contacts a reed switch have not broken in the same AC half-wave during which they have closed (10 ms)
$O_1^{'}$,$O_1^{''}$ the variables with values equal to the results of calculation of the 1st and 7th terms in (10)

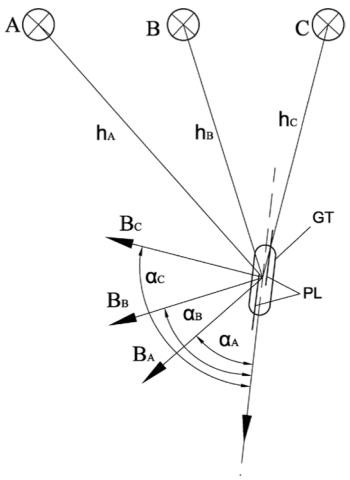


Fig. 1. Inductions of MFs which affect a reed switch (single column fitting image.).

frequency of 100 Hz and outputs signals in the form of rectangular pulses. Its action time t_{act} decreases [25] with an increase in the ratio $k = B_{lon}/B_{act}$, and $t_{act} \leq t_r$ (t_r is the rated time t_{act}) at $k \geq 1.5$. When a reed switch is located near a live conductor, the induction B_{lon} is produced by the current in this conductor and can be calculated by the Biot–Savart law. In three-phase current mains, a reed switch is affected by the sum of MFs with the inductions B_A, B_B, and B_C. These MFs are produced by currents I_A, I_B, and I_C in the phases A, B, and C of an electrical installation (Fig. 1). In this case,

$$B_{lon} = B_A\cos\alpha_A + B_B\cos\alpha_B + B_C\cos\alpha_C = \frac{\mu_0}{2\pi}\left(\frac{\cos\alpha_A}{h_A}I_A + \frac{\cos\alpha_B}{h_B}I_B + \frac{\cos\alpha_C}{h_C}I_C\right) \quad (1)$$

where α_A, α_B, and α_C are the angles between the longitudinal axis of the reed switch and the induction vectors B_A, B_B, and B_C; h_A, h_B, and h_C are the distances between the axes of the phases A, B, and C and the center of gravity of the reed switch.

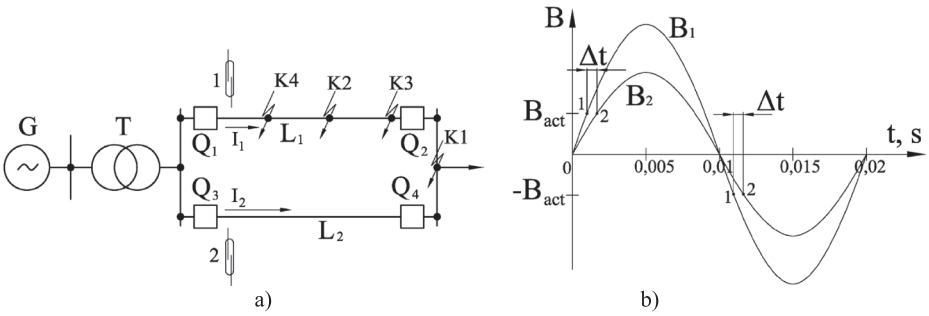


Fig. 2. (a) Circuit of a network section with double-circuit lines; (b) time of actuation of reed switches 1 and 2 in the case of a SC in the first line. (single column fitting image.).

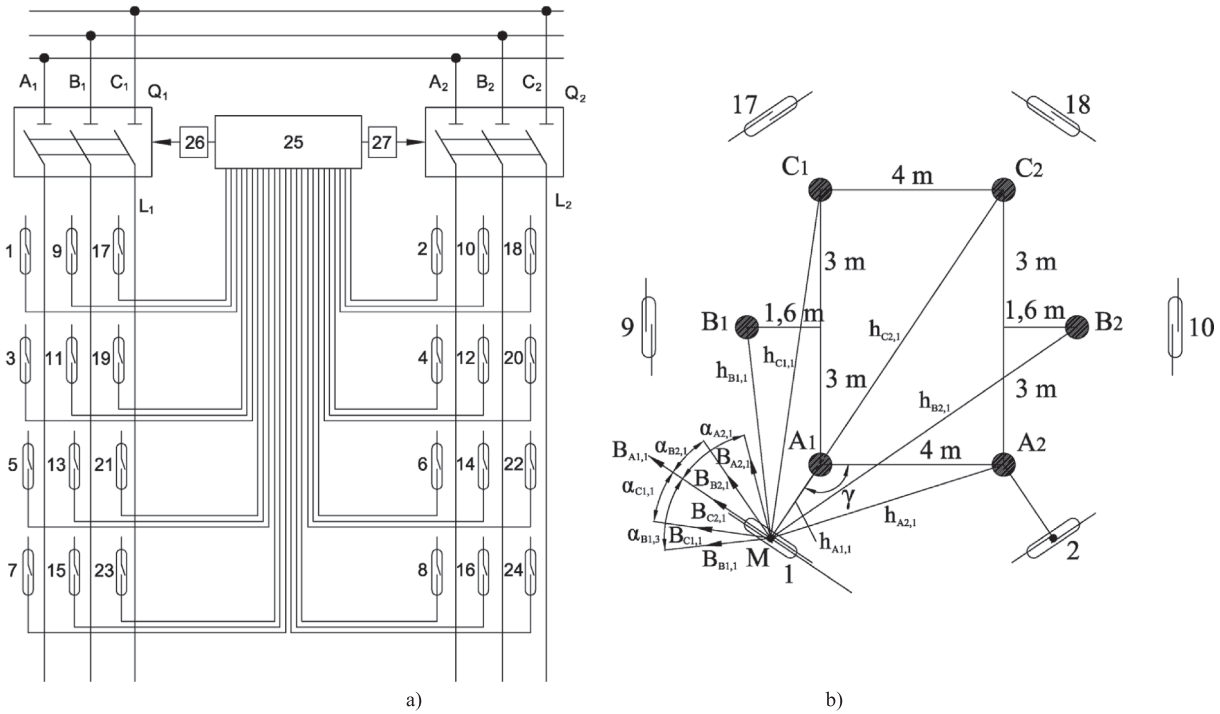


Fig. 3. (a) Protection device and (b) arrangement of the reed switches near the phases of the lines: reed switches (1–24); microcontroller (25); output elements (26 and 27). (single column fitting image.).

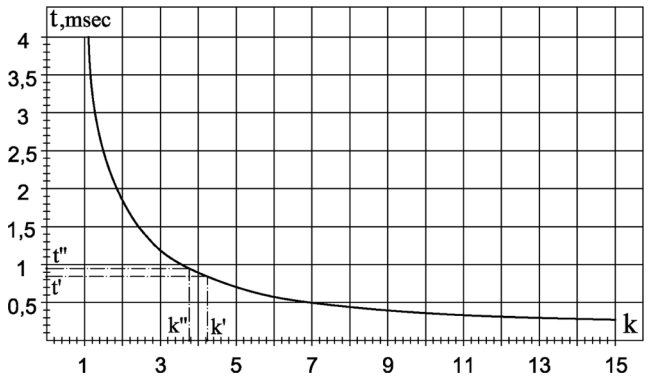


Fig. 4. Dependence of the time t from the B_{lon} zero-crossing time to the reed switch actuation on the ratio B_{lon}/B_{act} (1.5-column fitting image.).

3. Principle of protection operation

Principle of protection operation is based on the control of the time Δt between actuation of reed switches 1 and 2 fastened near the same phases of the lines (Fig. 2) and of the sequence of their actuation [26]. They are configured so as to actuate (close contacts) under the same currents in the lines. The time Δt is a kind of analogue of the differential current in a transverse differential directional protection. The Δt value determines whether a line short circuit (SC) occurs or not; and the sequence of actuation of the reed switches determines a fault line.

If reed switch 1 (2) actuates first, then a short circuit is in line L_1 (L_2). Let us show this. In the case of an external fault (point $K1$ in Fig. 2a), the currents I_1 and I_2 in the lines are equal (measurement and calculation errors are not taken into account for a while), and MFs with the same inductions B_1 and B_2 act on the reed switches. Therefore, they close the contacts at the same time, $\Delta t = 0$, and the protection does not operate. In the case of SC, for example, in line L_1 (point $K2$), $I_1 > I_2$ and $B_1 > B_2$ (Fig. 2b). Reed switch 1 actuates the first (point 1 in Fig. 2b) and signals, thus starting the time Δt reading until reed switch 2 actuation (point 2 in Fig. 2b). The time Δt is determined after its actuation. Since $\Delta t > 0$ and there is a signal from reed switch 1, then an opening signal is applied to breaker Q_1 of line L_1.

4. Discriminating elements of the protection

These are reed switches fixed at a safe distance from phases: reed switches 1, 3, 5, 7 for the phase A of line L_1, and reed switches 2, 4, 6, 8, of line L_2 (Fig. 3a); the same is for the phases B and C. Reed switches 1 and 2 are used to detect a fault line and measure the time Δt at $k_1 \le 1.5$ (k_1 is the ratio of the maximum to minimum SC current at the boundary of the cascade action zone), while reed switches 3 and 4, at $1.5 < k_1 \le 2$. Reed switches 5 and 6 (7 and 8) are used to block the protection under certain conditions, considered below. Any of the reed switches is affected by the sum of MFs produced by the currents in the phases A, B, and C of the first and second lines.

For reed switches 1 and 2, 3 and 4, 5 and 6, and 7, 8 to operate under the same currents in the lines, they are fasted symmetrical about the support (Fig. 3b) and should be of the same type. One of the simple ways to determine the point for fastening a reed switch is to fasten it tangentially to a circle (with the radius equal to the minimum allowable safety distance) circumscribed about the phase axis. In this case, it is affected by the maximal induction from the current in this phase, and we need only to find the point M (Fig. 3b) on this circle, where the total induction B^n_{lon}, which affects a reed switch, produced by the currents in the remaining five phases is minimal. The induction B^n_{lon} is calculated by Eq. (1), but five terms are used, which correspond to the phases where the currents interfere. The reed switch is mounted at this point.

5. Selection of the protection operation parameters

5.1. Selection of the time Δt_{th} between the instants of closing the contacts of reed switches 1 and 2 required for the protection operation

We mention above that $\Delta t = 0$ in the case of an external fault when errors are ignored. However, actually, $\Delta t = \Delta t_{im}$ (Δt_{im} is the time imbalance, by analogy with the imbalance current of differential protection) due to errors. The error ε_1 appears when a reed switch is mounted at the target point, the error ε_2 is due to the difference between the reed switch actuation times; and the error ε_3 is due to the deviation of B_{act} from the calculated value. Therefore, the protection should operate at the setting $\Delta t_{th} > \Delta t_{im}$.

To determine Δt_{im}, it is necessary to determine the dependence of the time t from the B_{lon} zero-crossing time to the reed switch actuation on the ratio $k = B_{lon}/B_{act}$. It is experimentally determined for each reed switch, for example, as shown in Appendix A. Fig. 4 shows this dependence for HSA-12126 reed switches with the rated action time $t_r = 0.3$ ms. We determine Δt_{im} when using them in the both lines.

Let $\varepsilon_1 = \pm 0.03$ [21], $\varepsilon_2 = \pm 0.1$, $t_r = 0.3$ ms [27], $\varepsilon_3 = \pm 0.03$; $k_2 = B_{lon}/B_{th1} = 4$ (B_{th1} is the induction under which reed switches 1 and 2 should actuate) in the case of a SC at point $K1$. Due to the error ε_1, reed switches 1 and 2 can be affected by the inductions $B'_{lon} = 1.03B_{lon}$ and $B''_{lon} = 0.97B_{lon}$. Since $\varepsilon_3 = 0.03$, one reed switch can actuate under $B'_{act} = 0.97B_{th1}$, and another, under $B''_{act} = 1.03B_{th1}$ in the worst case. The ratio k is $k' = B'_{lon}/B'_{act} = 1.03B_{lon}/0.97B_{th1} = 4.24$ and $k'' = B''_{lon}/B''_{act} = 3.76$. Using Fig. 4, we find the times $t' = 0.85$ ms and $t'' = 0.5$ ms corresponding to k' and $k\{\backslash Prime\}$. As a result, we find

$$\Delta t_{im} = t'' - t' + 0.2t_r = 0.95 - 0.85 + 0.2{\cdot}0.3 = 0.16 \text{ ms}$$

The time Δt_{im} is determined in the same way with other values of k_2 in the case of an external fault (point $K1$); Δt_{im} increases with a decrease in k_2. Therefore, if an SC occurs in the end of the line outgoing from busbars of a receiving substation, Δt_{im} can be larger than in the case of a busbar SC. To avoid the offset from this Δt_{im}, the protection is blocked if reed switches 5 and 6, which close contacts at $B_{th2} > B_{th1}$, does not actuate. Then,

$$\Delta t_{th} \ge 1.3\Delta t_{im1} \tag{2}$$

where Δt_{im1} is determined in the same way as Δt_{im}, but under the action of a MF with $B_{lon} = B_{th2}$ on reed switches 1 and 2; the offset coefficient 1.3 is responsible for the calculation and experimental errors, as is common in the relay protection.

5.2. Induction B_{th1} and B_{th2} under which reed switches 1, 2 and 5, 6 should actuate

As is shown in Appendixes B and C, the inductions B_{th1} and B_{th2} should satisfy the following conditions:

$$B_{th1} \le B_{th2}/1.5 \tag{3}$$

$$B_{th1} \ge 1.3\left(B_{load.\max} + B_{f1}\right) \tag{4}$$

$$B_{th1} \ge 1.3B_{f2} \tag{5}$$

$$B_{th2} \le B_{ca.r}/1.5 \tag{6}$$

$$B_{th2} \le B_{ca.\min}/1.5 \tag{7}$$

where $B_{ca.r}$ ($B_{ca.\min}$) is the induction amplitude of MF which acts on a reed switch in the minimal mode of system operation in the case of a double phase SC at the boundary of cascade action zone (section l_{ca} of the line from its opposite side, where $\Delta t < \Delta t_{th}$ under a SC) before circuit breaker opening from the receiving side (at busbars of the receiving part

in the case of double phase SC after the breaker opening); the required protection sensitivity factor $k_s = 1.5$ in case of SC in l_{ca}, like in traditional protections [28]; $B_{load.\max}$ is the induction amplitude of MF produced by the full-load current in a line; B_{f1} is the induction amplitude of MF acting on a reed switch near the phase of a fault-free line under the current in a fault line after circuit breaker opening from the side of the receiving substation in the case of SC in l_{ca} (calculated for a three-phase SC at the l_{ca} boundary in the maximal mode of system operation); B_{f2} is the induction amplitude of MF produced by the double earth fault current in the phases of the lines after opening the fault line circuit breaker from the receiving side.

Condition (3) ensures $\Delta t > \Delta t_{th}$ in the case of a SC at the l_{ca} boundary ($l_{ca} \leq 0.25\ l$ [28], l is the line length); conditions (4) and (5) ensure failure of the protection to break a fault-free line in the case of a SC near the busbars of the power supply substation and in the cascade zone, as well as in the case of a ground SC at two points of the network, one in the zone l_{ca} in the phase of the line protected and the other in the phase of the connection outgoing from the busbars of the receiving substation. Conditions (6) and (7) ensure the sensitivity of reed switches 5 and 6 to short circuit in l_{ca}, when $X_{S.\max} > 0.4X_L$ and $X_{S.\max} \leq 0.4X_L$, respectively ($X_{S.\max}$ is the maximal resistance of the system before the busbars the considered lines are connected to; X_L is the resistance of a line).

5.3 Time Δt_{th1} between the instants of closing contacts of reed switches 3 and 4, required for the detection of line SC and inductions B_{th3} and B_{th4} under which reed switches 3, 4 and 7, 8 should actuate

These reed switches and elements connected to them form an auxiliary part of the protection used to detect SCs in the lines protected before l_{ca} at $1.5 < k_1 \leq 2$. To avoid Δt_{th1} offset from the maximal Δt_{im}, this part is blocked if reed switches 7 and 8 do not actuate (like in the case of reed switches 1, 2 and 5, 6). In this case, for $\Delta t_{th1} = \Delta t_{th}$, it is convenient to take $B_{th4}/B_{th3} = B_{th2}/B_{th1}$. The B_{th4} value is limited by the need to ensure $k_s \geq 1.5$ at $B_{lon} > 1.5B_{ca.r}$, since reed switches 7 and 8 should actuate only when $k_1 > 1.5$. Hence,

$$B_{th3} \leq B_{th4}/1.5 \tag{8}$$

$$B_{th4} \leq B_{ca.r} \tag{9}$$

When $X_{S.\max} > 0.4X_L$, one can avoid mounting reed switches 3 and 4 and measure Δt between actuation of reed switches 5 and 6, because B_{th2} is selected from condition (6) and is equal to B_{th3}. Let us note that the required protection operation can be ensured (see justification in Appendix B), if $X_{S.\max} \geq 0.15X_L$, $\varepsilon_3 \leq 0{,}03$, and $t_r \leq 0.3$ ms for all the reed switches used for Δt measurement.

6. Protection circuit and algorithm of operation

Fig. 3a shows the line protection circuit, where $1.5 \leq k_1 \leq 2$. It includes reed switches 1–24, microcontroller 25, and output elements 26 and 27. Reed switches 1, 2, 9, 10, 17, and 18 actuate under B_{th1}; reed switches 3, 4, 11, 12, 19, and 20, under B_{th3}; reed switches 5, 6, 13, 14, 21, and 22, under B_{th2}, and reed switches 7, 8, 15, 16, 23, and 24, under B_{th4}.

Since many reed switches are used, we construct the protection so as it correctly operates if any of them fails. Malfunctions include: failure to operate, closed reed switch contacts for 10 ms after its actuation (the duration of the closed state of the contacts of a good reed switch is always less than 10 ms). Let us formulate the protection operation conditions in the case of SC on line L1 in terms of the boolean algebra. We will use boolean operators AND, OR, and NOT and the following designations: O_1 and O_2 for the trip signals to breakers Q_1 and Q_2 (Fig. 3a); G_i for the reed switch actuation signal; F_i for the operation failure signal; Z_i for the signal that the reed switch contacts remain closed for 10 ms after its actuation (i is the reed switch number in Fig. 3a); R_1 (R_2), R_3 (R_4), R_9 (R_{10}), R_{11} (R_{12}), R_{17} (R_{18}), and R_{19} (R_{20}) for the signals about actuation of reed switches 1 (2), 3 (4), 9 (10), 11 (12), 17 (18), and 19 (20) before reed switches 2 (1), 4 (3), 10 (9), 12 (11), 18 (17), and 20 (19), respectively; T_1, T_2, T_3, T_4, T_5, and T_6 for the signals that the time between the actuations of reed switches 1 and 2, 3 and 4, 9 and 10, 11 and 12, 17 and 18, and 19 and 20 attains Δt_{po} (2); T7 (T8) for the signal that 5 ms has elapsed since the actuation of reed switches 5, 13, or 21 (6, 14, or 22) (a good reed switch actuates before the induction attains the amplitude value, which takes 5 ms). All signals are boolean variables. If there is a signal, then the corresponding variable asserts, else deasserts (for the sake of brevity, we will omit the words “signal”, “logical zero”, and “logical unit”, but keep this in mind).

The first mode is a short circuit in section L1 before the cascade action zone l_{ca}. In this case, the currents on L1 are greater than the currents in L2; B_{po2} acts on the reed switches near L1, B_{po1} acts on the reed switches near L2, and $B_{po2} > B_{po1}$. Signal O1 is produced, if phase A is damaged and there are G1 AND G2 AND G5 AND there are R1 (reed switch 1 has actuated before reed switch 2) AND T1 (Δt between the actuations of reed switches 1 and 2 attains Δt_{po} (2)), which are stored for a time of 5 ms (time of 5 ms is taken with a margin based on the fact that all reed switches operate until the induction attains its amplitude value, which occurs in 5 ms after it passes through zero), AND there are no Z_1 AND Z_5 (prevent false tripping of L1 under an external fault, see explanations in Appendix D). Memorization (operator P^t) of signals R_1 and T_1 ensures the protection operation in the case of a short circuit on L1, otherwise it is impossible to determine the sequence of actuations of reed switches 1 and 2 and measure the time Δt, since reed switch 5 actuates after reed switches 1 and 2 ($B_{po2} > B_{po1}$), and signals from reed switches 1 and 2 simultaneously arrive at the time of closing the contacts of reed switch 5.

The second mode is a short circuit in the section L1 before l_{ca}, when the reed switches are affected by MFs with inductions B_{po4} near L1 and with inductions B_{po3} and near L2. Signal O_1 is produced if there are G_3 AND G_4 AND G_7 AND there are R_3 (reed switch 3 has actuated before reed switch 4) AND T_2 (the time between their actuations attains Δt_{po} (2)), which are stored for 5 ms, AND there are no Z_3 AND Z_7 (perform the same role as Z_1 and Z_5). The conditions for the protection operation of in these modes in the case of damage in phase B or C of line L1 are similarly formulated.

The third mode is a short circuit in the cascade action zone l_{ca}. Signal O_1 is produced if there is G_5 AND there is no Z_5 OR there is G_{13} AND there is no Z_{13} OR there is G_{21} AND there is no Z_{21} AND there is T_7 (5 ms has passed after the actuation of any of the reed switches 5, 13, or 21; 5 ms is sufficient to failure of actuation of reed switches 2, 10, and 18) AND, OR there are no G_2 AND G_{10} AND G_{18} AND F_2 AND F_{10} AND F_{18} OR there are no G_2 AND G_{10} AND G_{18} AND G_6 AND there is F_2, OR there are no G_2 AND G_{10} AND G_{18} AND G_{14} AND there is F_{10} OR there are no G_2 AND G_{10} AND G_{18} AND G_{22} AND there is F_{18} OR there are no G_{10} AND G_{18} AND G_6 AND there is G_2 AND Z_2 OR there is no G_2 AND G_{18} AND G_{14} AND there is G_{10} AND Z_{10} OR there are no G_2 AND G_{10} AND G_{22} AND there are G_{18} AND Z_{18}. Here, the signals Z_2, Z_{10}, Z_{18} and F_2, F_{10}, F_{18} are needed to avoid failure in the protection operation in this mode, and F_2, F_{10}, and F_{18} are also required to avoid a false L1 tripping signal in the case of a ground fault at two points, where one point is located on L2 in l_{ca} zone, and the second point is on the connection branching off from the busbars of the receiving substation; Z_5, Z_{13}, and Z_{21} are required to prevent tripping of L1 in the load mode after clearance of the external fault. The same condition is used to detect a short circuit on L1, when the currents in L2 produce an MF with the inductions less than B_{po1}. The protection operation conditions for opening circuit breaker Q2 (signal O2) in the above modes are similarly formulated.

Let us write the protection operation conditions in the analytical form according to the above formulations (the signs “+” and “·” denote boolean addition and multiplication, and the sign “–” above a symbol denotes boolean negation):

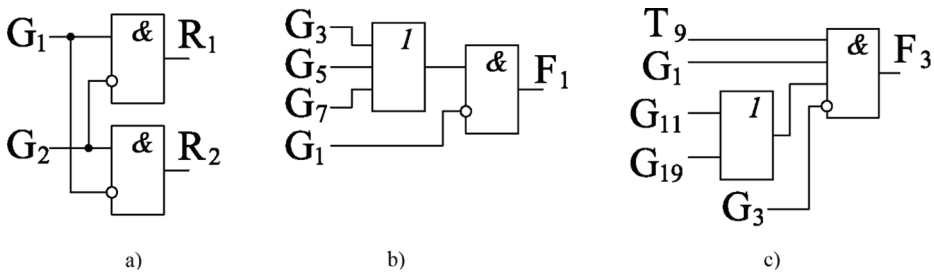


Fig. 5. Signal generation logic for R_1, R_2, F_1 and F_3.

$$\begin{gathered}O_1 = G_1{\cdot}G_2{\cdot}G_5{\cdot}(T_1{\cdot}R_1)P^t{\cdot}\overline{Z}_1{\cdot}\overline{Z}_5 + G_3{\cdot}G_4{\cdot}G_7{\cdot}(T_2{\cdot}R_3)P^t{\cdot}\overline{Z}_3{\cdot}\overline{Z}_7 + \\ +G_9{\cdot}G_{10}{\cdot}G_{13}{\cdot}(T_3{\cdot}R_9)P^t{\cdot}\overline{Z}_9{\cdot}\overline{Z}_{13} + G_{11}{\cdot}G_{12}{\cdot}G_{15}{\cdot}(T_4{\cdot}R_{11})P^t{\cdot}\overline{Z}_{11}{\cdot}\overline{Z}_{15} + \\ +G_{17}{\cdot}G_{18}{\cdot}G_{21}{\cdot}(T_5{\cdot}R_{17})P^t{\cdot}\overline{Z}_{17}{\cdot}\overline{Z}_{21} + G_{19}{\cdot}G_{20}{\cdot}G_{23}{\cdot}(T_6{\cdot}R_{19})P^t{\cdot}\overline{Z}_{19}{\cdot}\overline{Z}_{23} + \\ +(G_5{\cdot}\overline{Z}_5 + G_{13}{\cdot}\overline{Z}_{13} + G_{21}{\cdot}\overline{Z}_{21}){\cdot}T_7{\cdot}(\overline{G}_2{\cdot}\overline{F}_2{\cdot}\overline{G}_{10}{\cdot}\overline{F}_{10}{\cdot}\overline{G}_{18}{\cdot}\overline{F}_{18} + \\ +\overline{G}_2{\cdot}\overline{G}_{10}{\cdot}\overline{G}_{18}{\cdot}\overline{G}_6{\cdot}F_2 + \overline{G}_2{\cdot}\overline{G}_{10}{\cdot}\overline{G}_{18}{\cdot}\overline{G}_{14}{\cdot}F_{10} + \overline{G}_2{\cdot}\overline{G}_{10}{\cdot}\overline{G}_{18}{\cdot}\overline{G}_{22}{\cdot}F_{18} + \\ +\overline{G}_{10}{\cdot}\overline{G}_{18}{\cdot}\overline{G}_6{\cdot}G_2{\cdot}Z_2 + \overline{G}_2{\cdot}\overline{G}_{18}{\cdot}\overline{G}_{14}{\cdot}Z_{10}{\cdot}G_{10} + \overline{G}_2{\cdot}\overline{G}_{10}{\cdot}\overline{G}_{22}{\cdot}Z_{18}{\cdot}G_{18})\end{gathered} \tag{10}$$

$$\begin{gathered}O_2 = G_1{\cdot}G_2{\cdot}G_6{\cdot}(T_1{\cdot}R_2)P^t{\cdot}\overline{Z}_2{\cdot}\overline{Z}_6 + G_3{\cdot}G_4{\cdot}G_8{\cdot}(T_2{\cdot}R_4)P^t{\cdot}\overline{Z}_4{\cdot}\overline{Z}_8 + \\ +G_9{\cdot}G_{10}{\cdot}G_{14}{\cdot}(T_3{\cdot}R_{10})P^t{\cdot}\overline{Z}_{10}{\cdot}\overline{Z}_{14} + G_{11}{\cdot}G_{12}{\cdot}G_{16}{\cdot}(T_4{\cdot}R_{12})P^t{\cdot}\overline{Z}_{12}{\cdot}\overline{Z}_{16} + \\ +G_{17}{\cdot}G_{18}{\cdot}G_{22}{\cdot}(T_5{\cdot}R_{18})P^t{\cdot}\overline{Z}_{18}{\cdot}\overline{Z}_{22} + G_{19}{\cdot}G_{20}{\cdot}G_{24}{\cdot}(T_6{\cdot}R_{20})P^t{\cdot}\overline{Z}_{20}{\cdot}\overline{Z}_{24} + \\ +(G_6{\cdot}\overline{Z}_6 + G_{14}{\cdot}\overline{Z}_{14} + G_{22}{\cdot}\overline{Z}_{22}){\cdot}T_8{\cdot}(\overline{G}_1{\cdot}\overline{F}_1{\cdot}\overline{G}_9{\cdot}\overline{F}_9{\cdot}\overline{G}_{17}{\cdot}\overline{F}_{17} + \\ +\overline{G}_1{\cdot}\overline{G}_9{\cdot}\overline{G}_{17}{\cdot}\overline{G}_5{\cdot}F_1 + \overline{G}_1{\cdot}\overline{G}_9{\cdot}\overline{G}_{17}{\cdot}\overline{G}_{13}{\cdot}F_9 + \overline{G}_1{\cdot}\overline{G}_9{\cdot}\overline{G}_{17}{\cdot}\overline{G}_{21}{\cdot}F_{17} + \\ +\overline{G}_9{\cdot}\overline{G}_{17}{\cdot}\overline{G}_5{\cdot}G_1{\cdot}Z_1 + \overline{G}_1{\cdot}\overline{G}_{17}{\cdot}\overline{G}_{13}{\cdot}Z_9{\cdot}G_9 + \overline{G}_1{\cdot}\overline{G}_9{\cdot}\overline{G}_{21}{\cdot}Z_{17}{\cdot}G_{17})\end{gathered} \tag{11}$$

There are no the 2nd, 4th, and 6th terms in (10) and (11) for lines where $k_1 \le 1.5$. Signals about the failure of reed switches 3–8, 11–16, and 19–4 to operate are not used as well in (10) and (11), since they do not impact the correct operation of the protection when reed switches 1, 2, 9, 10, 17, and 18 are healthy. All signals, except for G_1–G_{24}, are generated inside microcontroller 25. For example, signals T_1–T_8 are produced by timers; signals R_1 and R_2, by the logic circuit (Fig. 5a, where 1, 2, 3, 5, 7, 11, 13, 15, 19, 21, and 23 are the contacts of the corresponding reed switches). Signals R_3, R_4, R_9–R_{12}, and R_{17}–R_{20} are generated in the same way as R_1 and R_2. Block diagrams of algorithms that implement the 1st (terms 2–6 are similar to the first one) and the 7th terms from (10) in microcontroller 25 are presented in Appendix E.

7. Reliability of the protection reed switches

The conditions for detection of reed switch failed to operated, e.g., 1, 3, 5, and 7, are formulates as follows. Signal F1 about the failure of reed switch 1 to operate is produced if there is no signal G_1 from this reed switch AND there is G_3 OR G_5 OR G_7. Signal F_3 about the failure of reed switch 3 to operate is produced if there is no G_3 AND there are G_1 AND T_9 (5 ms have passed since the actuation of reed switch 1) AND there is G_{11} OR G_{19}. Signal F_5 about the failure of reed switch 5 to operate is generated if there is no G_5 AND there is G_1 OR G_3 AND there is T_9 AND there is G_{13} OR G_{21}. Signal F_7 about the failure of reed switch 7 to operate is generated if there is no G_7 AND there is G_1 OR G_3 OR G_5 AND there is T_9 AND there is G_{15} OR G_{23}.

These conditions can be analytically written as

$$F_1 = \overline{G}_1{\cdot}(G_3 + G_5 + G_7); \tag{12}$$

$$F_3 = (G_1 + G_5){\cdot}\overline{G}_3{\cdot}T_9{\cdot}(G_{11} + G_{19}) \tag{13}$$

$$F_5 = G_1{\cdot}\overline{G}_5{\cdot}T_9{\cdot}(G_{13} + G_{21}) \tag{14}$$

$$F_7 = (G_1 + G_3 + G_5){\cdot}\overline{G}_7{\cdot}T_9{\cdot}(G_{15} + G_{23}) \tag{15}$$

Signals F_1, F_3, F_5, and F_7 are generated in microcontroller 25 according to the logic diagrams in Fig. 5b and 5c. The block diagram of algorithms (12) and (13) is given in Appendix E. The failures of the remaining reed switches to operate are detected in the same way, but also only in SC modes when they operate.

8. Protection operation in different modes

Let a three-phase SC occurs on busbars of the receiving substation, and SC currents in lines produce MFs with the inductions B_{po4}. If reed switches 1–24 (Fig. 3a) are good, then signals G_1–G_{24} about their actuation arrive to the inputs of microcontroller 25. Due to the errors, signals from reed switches 1 and 2, 3 and 4, 9 and 10, 11 and 12, 17 and 18, and 19 and 20 fixed near the same phases of lines L1 and L2 arrive at different times. But the time between their actuations $\Delta t < \Delta t_{po}$ (2); therefore no signals T_1–T_6. As a result, the first six terms in (10) and (11) are equal to zero. The 7th is also zero, since in contains inverted variables. As a result, $O_1 = O_2 = 0$ and the protection does not operate.

Let, in the case of a three-phase short circuit in the L1 section before l_{ca}, the currents in it and in line L2 produce MFs with the inductions B_{po4} and B_{po3}, respectively. Reed switches 3, 4, 11, 12, 19, and 20 actuates and produce signals G_3, G_4, G_{11}, G_{12}, G_{19}, and G_{20}. Since the currents in L1 are greater than the currents in L2, reed switches 3, 11, and 19 close contacts earlier than reed switches 4, 12, and 20, and signals R_3, R_{11}, and R_{19} are generated. The time between the actuations of reed switches 3 and 4, 11 and 12, and 19 and 20 attains Δt_{po}, and signals T_2, T_4, and T_6 are generated, respectively. The both groups of signals are memorized for 5 ms (P^t operator indicates this in (10)). Reed switches 7, 15, and 23 also actuate, but no>5 ms later than reed switches 3, 4, 11, 12, 19, and 20, since they react to the high induction value. Therefore, there are G_7, G_{15}, and G_{23}. There are no Z_3, Z_7, Z_{11}, Z_{15}, Z_{19}, and Z_{23}, since 10 ms has not elapsed after the actuation of reed switches 3, 7, 11, 15, 19, and 23. Therefore, the 2nd, 4th, and 6th terms in (10) are equal to 1, and $O_1 = 1$. Line L1 is tripped. In case of failure of, e.g., reed switch 15 to operate, the 4th term in (10) is equal to zero, since there is no G_7. But the 2nd and 6th terms are equal to 1. Therefore, $O_1 = 1$. The protection behaves similarly in the case of failure of other reed switches to operate.

In the case of a SC between, for example, the phases A and B in L_1 in l_{ca}, the protection does not produce signal O_1 to open Q_1 until opening the breaker on the receiving side. This can be explained by the following. Let the short-circuit currents in $L1$ and $L2$ produce MFs with the inductions higher than B_{po4}. Then, there are signals G_1–G_8 and G_9–G_{16} from reed switches 1–8 and 9–16, and the 7th term in (10) is zero. Terms 1–6 are also zero, since the difference between the currents in $L1$ and $L2$ is negligible under a short circuit in l_{ca}. Therefore, $\Delta t < \Delta t_{po}$, there are no signals T_1–T_6, and $O_1 = 0$. After tripping $L1$ from the receiving side, the currents in the phases of $L2$ are insufficient (the entire SC current flows through $L1$) for actuation of reed switches 2, 4, 6, 8, 10, 12, 14, and 16. Therefore, $G_2 = G_4 = G_6 = G_8 = G_{10} = G_{12} = G_{14} = G_{16} = 0$, and signals F_2 and F_{10} are also absent (they are generated similarly to F_1 following (12)), since reed switches 2 and 10 are in good order. Reed switches 5, 13, and 21 signal, $G_5 = G_{13} = G_{21} = 1$, but there are no Z_5, Z_{13}, and Z_{21}, because 10 ms has not elapsed after their actuation. As a result, the 7th term in (10) is equal to 1, and $O_1 = 1$. If, for example, reed switch 2 has not actuated before tripping $L1$ from the receiving side due to a fault, then signal $G_2 = 0$ after tripping $L1$, and $F_2 = 1$. Therefore, the 7th term is equal to 1 and $O_1 = 1$. If reed switch 2 has actuated and its contacts remain closed, then $G_2 = 1$ and $Z_2 = 1$, since the contacts of reed switch 2 have not broken within 10 ms. But reed switch 6 does not actuate ($G_6 =$

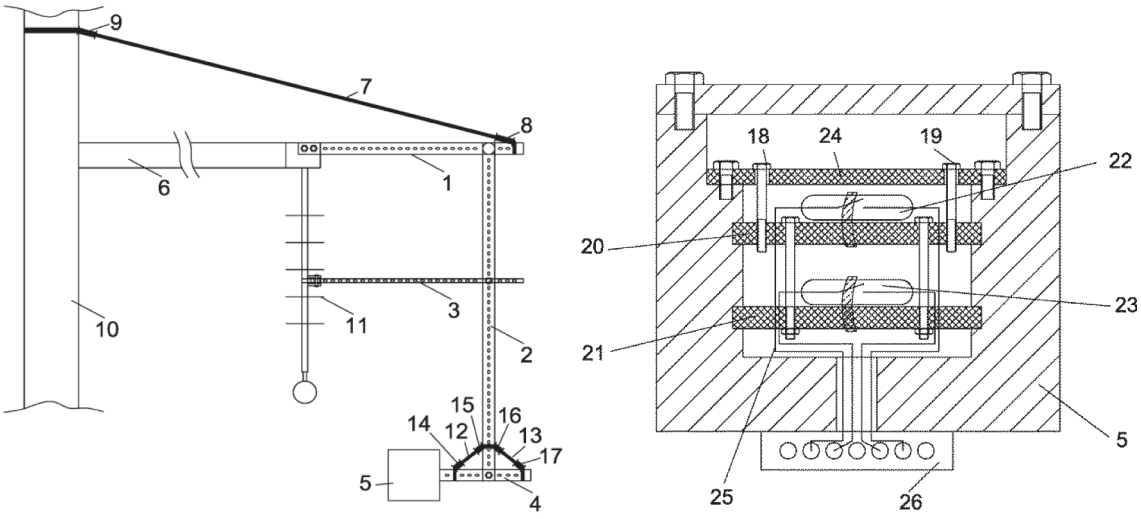


Fig. 6. (a) General view of the structure for fastening reed switches; (b) case 5 in section (single column fitting image.).

0). As a result, $O_1 = 1$.

Let us consider a ground fault at two points, where one point is in l_{ca}, for example, on phase A of line $L1$, and another is on phase B of the connection branching off from the busbars of the receiving substation. Let us show that the protection does not produce Q_1 and Q_2 trip signals. Before $L1$ is tripped from the receiving side, short-circuit currents flow through phases A and B of lines $L1$ and $L2$. In phases B they are equal, but in phases A they are different. Therefore, there are signals G_1–G_8 and G_9–G_{16}, if reed switches 1–8 and 9–16 are in good order, but the time between the instants of actuation of reed switches 1 and 2, 3 and 4, 9 and 10, and 11 and 12 is shorter than Δt_{po}, and there are no T_1, T_2, T_3, and T_4. Hence, $O_1 = O_2 = 0$, and $L1$ and $L2$ remain in operation. After tripping $L1$ from the receiving side, short-circuit currents flow through its phase A and phase B of line L2. Reed switches 1, 3, 5, 7, 10, 12, 14, and 16 actuate, and signals G_1, G_3, G_5, G_7, G_{10}, G_{12}, G_{14}, and G_{16} are produced, as well as signals R_1, R_3, R_{10}, and R_{12} (that reed switches 1, 3, 10, and 12 have actuated before reed switches 2, 4, 11, and 13). Therefore, all terms in (10) and (11) are equal to zero, and $O_1 = O_2 = 0$. If reed switch 1 did not close the contacts ($G_1 = 0$) or closed ($G_1 = 1$) and did not break within 10 ms ($Z_1 = 1$) because of a fault, then $O_2 = 0$, since there is G_5. Other modes are considered in a similar way.

9. Structure for fastening reed switches

She (Fig. 6) includes [29] bars 1–4 with scales and holes and case 5. One end of bar 1 is attached to traverse 6, and another end is attached to support 10 by cable 7 and tambuckles 8 and 9. The electrical safety distance is counted on the scale of bar 1, and it is fastened to one end of bar 2 at this point. Bar 3 is attached to set of insulators 11 at one end and to bar 2 at the other end so as bars 1 and 2 are mutually perpendicular. The distances from traverse 6 and the axis of bar 2 to the reed switch fixing point are measured on the scales of bars 2 and 4. Bars 2 and 4 are connected with a bolt. Bar 4 is fixed with cables 12 and 13 and tambuckles 14–17. Using bolts 18 and 19 as handles, rigidly fastened disks 20 and 21 are rotated in the grooves of case 5 to a required angle (counted on the scale on cover 24). Their position is fixed by screwing bolts 18 and 19 into the holes in disk 20. Reed switches 22 and 23 are connected to microprocessor 25 by cables through terminal block 26. When designing the protection with four reed switches per phase, another case 5 with two reed switches is attached to bar 4. Other reed switches are fastened at points specified using the same structure. The weight of each of them does not exceed 5 kg. Note that this structure can also be used for 6–10-kV lines.

Table 1
An example of a table.

Voltage (kV)	Weight (kg)			
	Total	Copper	Steel	Insulation
6	60/11	9/1	21/2	30/-
10	90/11	12/1	33/2	45/-
35	300/30	32/1.5	118/3.5	150/-

10. Estimated resource saving due to the protection suggested

Table 1 presents the weight and size parameters of the protection suggested and an ABB protection with six current transformers.

This table shows that the traditional protection requires 9 and 10 times more copper and steel, respectively, than the protection suggested at a voltage of 6 kV, 12 and 16 times more, at a voltage of 10 kV, and 21 and 33 times more, at a voltage of 35 kV. It also requires 30–150 kg of high-voltage insulation, which is not required for the protection suggested. The analysis of similar data for protections of several other companies has showed the ratios close to those given in Table 1.

11. Example of calculation of the protection operation parameters for lines in Fig. 2a

Transformer T is of TDN-40000/110 type with a low voltage of 38.5 kV. The voltage control range is ± 16%; the short-circuit voltages $u_{SC.max} = 11.02\%$ and $u_{SC.min} = 10.35\%$. The lines are made from AC 120/19 cable with the resistances $r_{spec} = 0.25$ Ohm/km and $X_{spec} = 0.41$ Ohm/km; their length is 30 km (PM35-2FT supports). The distances between the phases are given in Fig. 3b. A power of 8 MW is transmitted through every line at $cos\varphi = 0.9$. The system resistance before the transformer $X_S = 0$. The rated load current amplitude $I_{load.r} = 0.207$ kA.

Let us find the ratio between $X_{S.max}$ and X_L. $X_S = 0$; therefore, $X_{S.max} = X_{T.max} = 4.85$ Ohm ($X_{T.max}$ – maximal resistance of the transformer). Since $X_{S.max} > 0.15X_L$, we continue the calculations. In the case of a SC at the boundary $l_{ca} = 0.25\ l$, we calculate the maximum currents $I_{SC.L1} = 2.06$ kA and $I_{SC.L2} = 1.24$ kA in L_1 (fault) and L_2, respectively, and the minimum current $I_{1ca} = 1.46$ kA in L_1. They are required to determine the number of reed switches to be mounted near a phase and to check the performance of the protection. Since $k_1 = I_{SC.L1}/I_{1ca} = 1.4 < 1.5$, then

Table 2
Maximal length l (km) of lines the protection can be mounted at.

Cross section, mm^2	70	95	120	150	185	240
Rated power of the transformer (T), MVA	Length (l), km					
16	45	20	–	–	–	–
25	45	35	20		–	–
40	45	40	30	20	–	–
63	45	45	40	35	20	–
80	35	45	40	35	30	20

two reed switches per phase are to be mounted.

To calculate B_{f1}, we find the maximum current $I_{tr.L1} = 2.31$ kA in L_1 with a short circuit at the boundary l_{ca} (L_2 is broken), and to determine $B_{ca.\min}$, the minimum current $I_{ca.\min} = 1.41$ kA in the event of a bus SC in the receiving substation. To calculate B_{f2}, we find the current $I_{SC.ca}$ of an earth fault at two points. It is maximal when the first closing point is located at the boundary l_{ca}, and the second, near the busbars of the receiving substation. Taking into account that the double fault current is comparable with the current I_{SC} of a double phase SC, we take $I_{SC.ca} = I_{SC}$ to ensure the B_{th1} margin, which is determined in the case of a short circuit at the boundary l_{ca} in the maximum mode of the system operation. Then, $I_{SC.ca} = I_{SC} = 2.01$ kA.

Let us select the parameters of the protection operation with reed switches 1, 5, 2, and 6. The induction B^n_{lon} acting, for example, on reed switch 1 is minimal if the point M (Fig. 3b), where it is fixed, belongs to the plane under the phases, and $\gamma = 124^\circ$ (Fig. 3b). In this case, the cosines of the angles between the axis of reed switch 1 and inductions produced by currents in the phases A, B, and C of both lines are $\cos\alpha_{A1,1} = 1$, $\cos\alpha_{B1,1} = 0.77$, $\cos\alpha_{C1,1} = 0.9$, $\cos\alpha_{A2,1} = 0.74$, $\cos\alpha_{B2,1} = 0{,}93$, and $\cos\alpha_{C2,1} = 1$ (in the subscripts, the numbers 1 and 2 after the letters A, B, and C means the first and the second lines, and the number after a comma is the reed switch number). The distances between the point M and the axes of phases of the lines are $h_{A1,1} = 0.4$ m, $h_{B1,1} = 4.63$ m, $h_{C1,1} = 7.68$ m, $h_{A2,1} = 5.33$ m, $h_{B2,1} = 8.11$ m, and $h_{C2,1} = 9.14$ m. Reed switch 5 is fixed next to reed switch 1 at a point on a straight line passing through point M parallel to the phase A_1 at the same angle $\gamma = 124^\circ$. Therefore, the ratios of the cosines to the distances found for point M are also valid for the reed switch 5 point. Reed switches 2 and 6 are mounted symmetrically to reed switches 1 and 5 relative to the support.

Let us calculate the absolute values of the inductions $B_{load.\max}$, $B_{ca.\min}$, B_{f1}, and B_{f2} which act along the axes of reed switches 1 and 5:

$B_{load.\max} = 1.94 \times 10^{-4}$ T; $B_{ca.\min} = 6.62 \times 10^{-4}$ T; $B_{f1} = 1.83 \times 10^{-5}$ T; $B_{f2} = 1.71 \times 10^{-5}$ T.

Following (7) and (3), $B_{th2} = B_{ca.\min}/1.5 = 4.4 \times 10^{-4}$ T; $B_{th1} = B_{th2}/1.5 = 2.93 \times 10^{-4}$ T.

We take these induction values as operation settings, since conditions (4) and (5) is fulfilled, and the reed switches are selected based on them. Let reed switches 1, 2, 5, and 6 be of the HSA-12126 type, with the actuation inductions 3×10^{-4} T, 2.9×10^{-4} T, 4.3×10^{-4} T, and 4.5×10^{-4} T, respectively. In this case, $\Delta t_{th} = 460$ μs (Appendix B). Let us check whether the protection detects a short circuit, for example, in L_1, at the boundary of l_{ca}. We calculate the amplitudes of inductions $B_{SC.L1}$ and $B_{SC.L2}$ under currents $I_{SC.L1} = 2.06$ kA and $I_{SC.L2} = 1.24$ kA in L_1 and L_2 taking into account $\varepsilon_1 = 0.03$: $B_{SC.L1} = 9.4 \times 10^{-4}$ T and $B_{SC.L2} = 5.97 \times 10^{-4}$ T. We find the ratios $B_{SC.L1}/B_{th1} = 3.13$ and $B_{SC.L2}/B_{th1} = 2.06$. Using Fig. 4 we find the time $\Delta t = 650$ μs. The operation settings of the other reed switches and Δt between the actuation of reed switches 9 and 10, 17 and 18 are also determined. Similar calculations showed that the protection can be mounted at lines no longer then the values given in Table 2.

12. Results and discussion

The protection without current and voltage transformers is suggested for double-circuit transmission lines for the first time. It allows saving copper, steel, and high-voltage insulation in amounts unprecedented for relay protections. Thus, the consumption of these materials for 6–35-kV lines can be reduced by 8–30, 20–115, and 30–150 kg, respectively. The protection is quite simple: two reed switches are fixed near each phase and connected to a microprocessor, if $k_1 \leq 1.5$ (k_1 is the ratio of maximum to minimum SC current at the boundary of the cascade action zone), and four reed switches, if $1.5 < k_1 \leq 2$. A technique for selection of the protection operation parameters and an algorithm of its operation are developed. A design is suggested for fastening reed switches at the PM35-2FT support. It can be used with other supports with minor modifications.

The protection suggested meets the sensitivity requirements [28] for transverse directional protections if: (a) the operation inductances of the selected reed switches differ from the calculated values by no more than ± 3%; (b) the rated action time of the reed switches, between the time of closing the contacts of which the time is measured, does not exceed 0.3 ms. The protection is not inferior to up-to-date protection devices in terms of speed, since it is also based on a microprocessor, and the reed switches actuate within 5 ms from the instant of occurrence of a short circuit.

For reliable operation of the protection in all modes, the algorithm of its operation takes into account a probability of failure of any of the reed switches. A possibility of its detection is also provided, but only in the event of a short circuit. To do this in the load mode, it is necessary to equip the protection with test diagnostics– reed switches are used with a control winding, where signals are sent from an external voltage source simulating different short circuit types. Further, the protection operates following the above algorithms. To improve its reliability, the majorization principle can be used. For this, two similar protections are additionally mounted, and a fault line breaking signal is produced if two of the three protections have operated. Current sensors (reed switches) are also duplicated, unlike protections with current transformers. This increases its reliability in comparison with well-known non-microprocessor protections. However, only operating experience will show which design is more reliable.

As for the field of use of the protection, it is smaller than that of devices with current transformers. It is limited by a need to take into account the noise effect and a possibility of implementing the principle in those lines where the resistance of one circuit does not exceed the maximal resistance of the system (including the resistance of the supply transformer) by>6.6 times, and the ratio of maximal to minimal short-circuit current at the boundary of the cascade action zone does not exceed 2.

We should note that the reed switch differential protection [21] can be already used instead of classical protections with current transformers (CTs) at 10(35)-kV power transformers of at least 6.3 (40) MVA in power. The field of use of the protection suggested is presented in Table 2 and in the previous paragraph. Reed switch current protections [6] can be used for 6–110-kV electrical installations like traditional ones, except the cases where short-circuit currents are comparable with load currents. The protection reed switches (no more than three) should be provided with windings and connected to appropriately calibrated standard measuring instruments after increased the EMF induced on its terminals by the phase current. This is necessary because traditional overcurrent protections are often connected to the same CTs as the measuring instruments. The design of reed switch protections for electrical installations with voltages above 110 kV is still limited by insufficient sensitivity of reed switches due to large (according to safety regulations) distances from busbars with current. The special role of reed switch protections is that they can help to increase the reliability (by dozens of times theoretically) of the existing relay protection systems with CTs via duplication by the majorization principle. Current

transformers are also duplicated, but this is not done now, which is one of the reasons for the delay in using this principle.

13. Conclusions

Measuring the time between the actuation of two reed switches fixed near the same phases of double-circuit lines and sequencing their actuation we have designed a short-circuit protection without current and voltage transformers. It opens prospect for unprecedented resource saving in the relay protection of 6–35-kV double-circuit lines, i.e., up to 30, 110, and 150 kg of copper, steel, and high-voltage insulation, respectively. The protection operation algorithm described ensures fail-safety in the cases of failure of reed switches and signaling about them and can be implemented on the basis of logical components of any nature. The analysis of the microprocessor protection with a logical part which functions following this algorithm allows expecting its correct operation in all modes and sensitivity and speed no worse that those of traditional transverse directional protections. However, it should be used in the cases where the maximal resistance of the system (including the line supplying transformer) is at least 15% of the resistance of one line circuit at a rated reed switch action time of up to 0.3 ms.

Funding sources

This research is funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP13268753).

CRediT authorship contribution statement

M.Ya. Kletsel: Conceptualization, Methodology, Writing – review & editing. **B.E. Mashrapov:** Data curation, Writing – original draft, Funding acquisition. **R.M. Mashrapova:** Investigation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The main data is given in the article, the rest data will be made available on request

Appendix A. Experimental derivation of the dependence of the time t on the ratio k

To derive the dependence of t on k, experiments were carried with the setup shown in Fig. A.1. Four HSI Sensing reed switch types and six Russian reed switches with $t_r = 0.3$–2 ms were studied. Applying a DC voltage to induction coil (IC) 1 (with a reed switch fixed inside) from the first outputs of source 2 through breakers 3, the voltage U_1, under which the reed switch was activated, was recorded at the outputs of IC 1 by voltmeter 4 (U_1 is proportional to B_{act}). The closure of its contacts was determined using oscillograph 5 by rectangular voltage pulses at resistor 6 connected through these contacts to the second outputs of source 2. Then, the alternating voltage U_2 (proportional to B_{lon}) from source 9 was applied to IC 1 through breakers 7 and resistor 8. In that case, the ratio k of the amplitude of that voltage to U_1 varied from 1 to 17. Further, at those ratios, the time t was measured from the U_2 zero-crossing time to the reed switch actuation time by oscillograms of voltages measured at resistors 6 and 8.

Appendix B. Statement of the need for fulfilment conditions (3), (6), and (7)

To explain how Eq. (3) is derived, we determined B_{th2}. Let us begin from conditions (6) and (7). In the case of SC at the boundary of the zone $l_{ca} = 0.25\,l$, e.g., in L_1,

$$I_{1ca} + I_{2ca} = 0.87U/(X_{S.\max} + 0.47X_L) \tag{A.1}$$

$$I_{1ca}/I_{2ca} = (l + l_{ca})/(l - l_{ca}) = 1.66 \tag{A.2}$$

where U is the rated voltage of the lines; I_{1ca} and I_{2ca} are the currents in L_1 and L_2 in the case of a double phase SC at the boundary of l_{ca} in the minimal system operating mode.

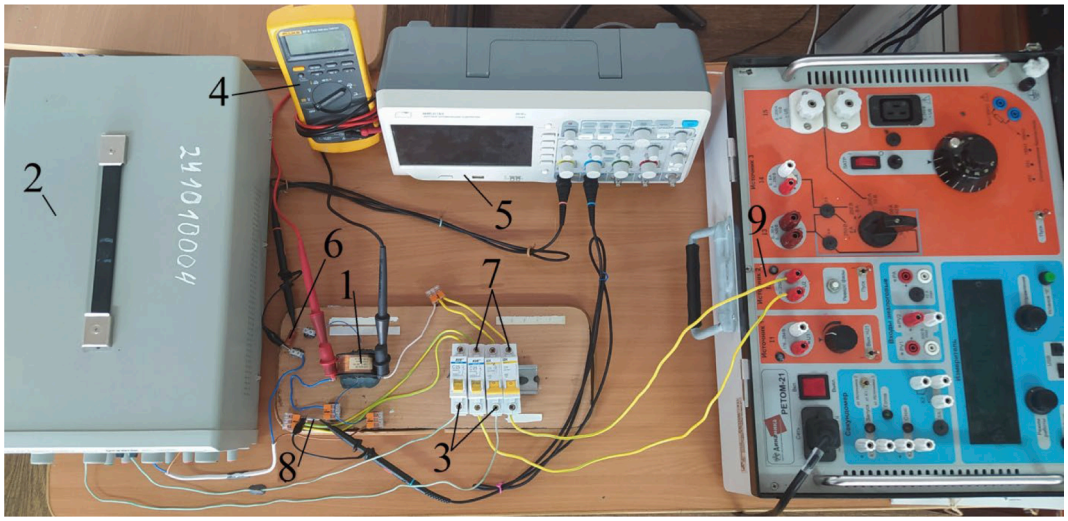


Fig. A.1. Laboratory setup (2-column fitting image.).

In the event of a bus SC in the receiving substation, when one of the lines is broken, the current

$$I_{ca.\min} = 0.87U/(X_{S.\max} + X_L) \tag{A.3}$$

From Eqs. (A1) - (A3),

$$I_{1ca}/I_{ca.\min} = 0.6(X_{S.\max} + X_L)/(X_{S.\max} + 0.47X_L); \tag{A.4}$$

The ratio is less than 1 when $X_{S.\max} > 0.4X_L$. Hence, the ratio of the inductions $B_{ca.r}$ and $B_{ca.\min}$ is the same. In the both cases, the sensitivity coefficient $k_s \geq 1.5$ is required. Therefore, condition (6) should be satisfied when $X_{S.\max} > 0.4X_L$, and condition (7), when $X_{S.\max} \leq 0.4X_L$.

Let us find the relationship between B_{th1} and B_{th2} to ensure $\Delta t > \Delta t_{th}$ in the case of a SC at the boundary $l_{ca} = 0.25\,l$. Let us analyze the values of Δt in this mode at different k_1 and compare them with Δt_{th} calculated at different $k_3 = B_{th2}/B_{th1}$ with allowance for the errors. Let $X_{S.\max}$ less than $0.4X_L$ and $k_3 = 1.5$. Then the maximum possible values $B_{th2} = 0.67B_{ca.\min}$ and $B_{th1} = 0.44B_{ca.\min}$. To determine the ratio of inductions acting on reed switches 1 and 2 in the case of a SC at the boundary of l_{ca}, we express them in terms of $B_{SC.\min}$ (i.e., the induction of a MF produced by the current $I_{SC.\min}$ in one line circuit in the case of a double phase bus SC in the receiving substation in the minimal system operating mode when both lines are live). The current $I_{SC.\min} = 0.87U/2(X_{S.\max} + 0.5X_L)$. Considering it and Eqs. (A1) – (A3), we derive

$$I_{2ca}/I_{SC.\min} = 0.75(X_{S.\max} + 0.5X_L)/(X_{S.\max} + 0.47X_L) \tag{A.5}$$

$$I_{2ca}/I_{SC.min} = 1.25(X_{S.max} + 0.5X_L)/(X_{S.max} + 0.47X_L) \tag{A.6}$$

$$I_{ca.\min}/I_{SC.\min} = (2X_{S.\max} + X_L)/(X_{S.\max} + X_L) \tag{A.7}$$

According to Eqs. (A5) – (A7), with, e.g., $X_{S.\max} = 0{,}15X_L$, the currents $I_{1ca} = 1.31I_{SC.\min}$, $I_{2ca} = 0.79I_{SC.\min}$, and $I_{ca.\min} = 1.13I_{SC.\min}$. Hence, the inductions of the MF produced by these currents $B_{ca.r} = 1.31B_{SC.\min}$, $B_{ca.n} = 0.79B_{SC.\min}$, and $B_{ca.\min} = 1.13B_{SC.\min}$; $B_{th1} = 0.5B_{SC.\min}$. Let $k_1 = 1.5$. Then, the inductions produced by the currents in the fault and fault-free lines in the maximal system operating mode $B_{ca1} = 1.97B_{SC.\min}$ and $B_{ca2} = 1.19B_{SC.\min}$. With allowance for $\varepsilon_1 = \varepsilon_3 = 0.03$, we can find the ratios

$$B_{ca.r}/B_{th1} = 1.32B_{SC.\min}(1-\varepsilon_1)/((1+\varepsilon_3)0.47B_{SC.\min}) = 2.47;$$

$$B_{ca.n}/B_{th1} = 0.8B_{SC.\min}(1+\varepsilon_1)/((1-\varepsilon_3)0.47B_{SC.\min}) = 1.68;$$

$$B_{ca1}/B_{th1} = 3.7$$

$$B_{ca2}/B_{th1} = 2.5$$

Using these ratios and Fig. 4, we find the time between actuation of reed switches 1 and 2 in the minimal and maximal system operation modes: $\Delta t_{SC.\min} = 0.74$ ms and $\Delta t_{SC.\max} = 0.48$ ms. Then, $\Delta t_{im1} = 0.35$ ms at $k_3 = 1.5$ and $\Delta t_{th} = 0.46$ ms. Since $\Delta t_{SC.\max} > \Delta t_{th}$, $l_{ca} \leq 0.25\,l$. The same calculations for other ratios k_1 and k_3 and $X_{S.\max}$ and ε_3 with the use of the dependences of t on k and for reed switches with $t_r > 0.3$ ms have shown the following.

(1) To ensure sensitivity, the reed switches used to measure Δt should have $t_r \leq 0.3$ ms, and $\varepsilon_3 \leq 0.03$ is required. For reed switches with $t_r > 0.3$ ms, the time t changes faster, and with $\varepsilon_3 > 0.03$, Δt_{th} is longer, which does not allow selecting k_1, k_3, and $X_{S.\max}$ at $l_{ca} \leq 0.25\,l$.
(2) At $k_1 \leq 1.5$, for $l_{ca} \leq 0.25\,l$, it is necessary to fix two reed switches near each phase, $k_3 = 1.5$, and $X_{S.\max} \geq 0.15X_L$.
(3) At $k_1 > 1.5$, for $l_{ca} \leq 0.25\,l$, it is necessary to ensure actuation of these reed switches at the inductions higher than B_{th1} and B_{th2}. It is difficult to change their actuation inductions versus k_1. Therefore, auxiliary reed switches are mounted near each phase. Thus, at $1.5 < k_1 \leq 2$, two more reed switches are mounted (3, 7 and 4, 8 in Fig. 3a); one actuates at $B_{th3} > B_{th1}$, and another, at $B_{th4} > B_{th2}$.
(4) If $k_1 > 2$, more reed switches should be mounted. This is unreasonable, since the protection becomes more complex and its reliability decreases.

Appendix C. Explanation of conditions (4) and (5)

Let us consider why conditions in (4) are required. Let a phase-to-phase short circuit occurs in L_1 in the l_{ca} zone. The protection does not operate before braking L_1 on the receiving side, since $\Delta t < \Delta t_{th}$. After the breaking, short-circuit current flows in L_1, and load current flows in L_2. Let B_{th1} does not satisfy (4), and the current in L_2 is sufficient for actuation of reed switch 2, but not reed switch 6 (Fig. 3a). The protection does not detect a short circuit, since reed switch 2 closes the contacts earlier than reed switch 1 (the load current leads the short circuit current in phase). In this case, the actuation of reed switch 5 and non-operation of reed switch 6 indicate a large difference between the currents in L1 and L2. This difference can take place in the case of an external fault because of errors. To distinguish between these modes, reed switch 2 should be offset from the maximal effective induction after breaking L_1 from the receiving side in the event of a short circuit in l_{ca}, i.e. (4) should fulfill. Then, a Q_1 opening signal is produced if one of the reed switches with B_{th2} near L_1 has actuated, and all the reed switches with B_{th1} near L_2 failed to actuate. The same is in case of a short circuit in L_1 in the vicinity of the busbars of the supply substation, if the currents in L_2 produce a MF with the induction lower than B_{th1}.

In the case of a ground fault at two points, let one of them be at the phase A1 and another be at the phase B of the connection connected to the busbars of the receiving substation. The protection does not operate before the circuit breaker opening from the receiving side, since $\Delta t < \Delta t_{th}$ (currents in the phases B_1 and B_2 are equal; and the phase A_1 short circuit point is in l_{ca}). After the breaker opening, reed switches 9 and 17 (Fig. 3a) near the phases B_1 and C_1 and reed switches 2 and 18 near the phases A_2 and C_2 do not actuate, since they are offset from the effect of currents in the phases A_1 and B_2 according to (5). As a result, Δt is not measured. The conditions for protection operation considered in the case of a phase-to-phase short circuit in l_{ca} are not satisfied as well. Therefore, no Q_1 and Q_2 opening signals are produced.

Appendix D. Explanation of the use of Z_1 and Z_5 signals

In the external fault mode, reed switches 1, 2, 5, and 6 are affected by equal inductions (equal currents in $L1$ and $L2$). Reed switches 1 and 2 close the contacts the first, because the induction of their actuation is lower than that of reed switches 5 and 6. At the time of actuation (point 1 in Fig. D.1),

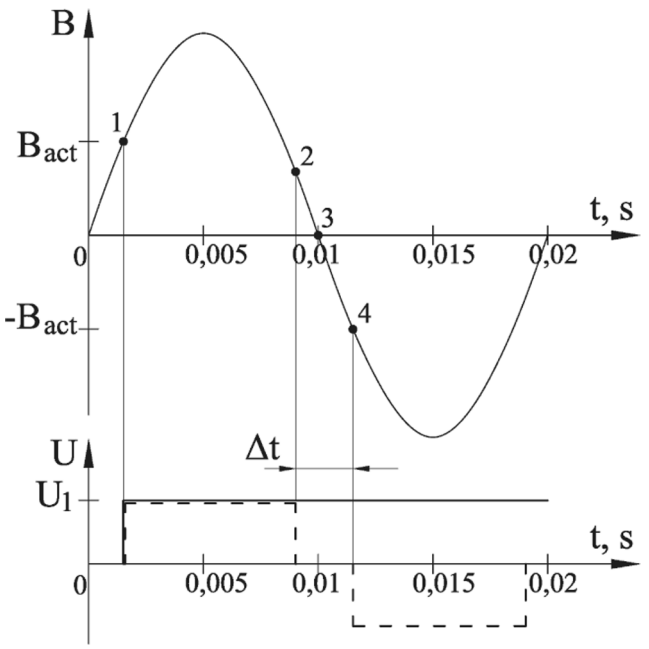


Fig. D.1. Time $\Delta t = 10$ ms within which the contacts of reed switch 1 are closed and the contacts of reed switch 2 break (the signal from reed switch 1 is shown by the solid curve, and from reed switch 2, by the dashed curve).

they send signals G_1 and G_2 (voltages U_1 are shown by the solid and dashed curves) to different inputs of microcontroller 25. Within 5 ms after reed switches 1 and 2 is operate, reed switches 5 and 6 close the contacts. If reed switches 1 and 2, 5 and 6 are in working order, then they stop signaling (signals G_1 and G_2 at point 2) before the induction sine curve crosses zero (point 3), and the signal duration does not exceed 10 ms. Therefore, there are no Z_1 and Z_5. If the contacts of reed switch 1 do not return to the original state, and of reed switch 2, return, then signal G_1 continues (the solid curve in Fig. D.1 does not break at point 2) and signal G_2 ceases. Therefore, at the time of breaking the contacts of reed switch 2 (point 2), signal R_1 is generated (that reed switch 1 has actuated the first), and the time Δt is started counting until their re-closing (point 4). In this case, $\Delta t > \Delta t_{po}$, and signal T_1 is produced. Signals R_1 and T_1 are memorized for 5 ms. Reed switch 5 closes the contacts within the same time and produces signal G_5. However, the protection does not produce signal O_1, since 10 ms elapses before the actuation of reed switch 5, during which the contacts of reed switch 1 are closed, and signal Z_1 is produced.

Appendix E. Block diagrams of algorithms

Fig. E.1 shows the block diagrams of algorithms for implementation of the 1st and 7th terms in (10) in the microcontroller. Fig. E.2 shows the block diagrams of algorithms for detecting malfunctions of reed switches 1 and 3. In the block diagrams: U_i (i = 1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 18, 19, 21, 22) is the voltage applied to the microcontroller input when the contacts of the corresponding reed switch are closed; U_s is the voltage at which the microcontroller considers a signal to arrive at its input; G_i, Z_i, R_i, and F_i are the signals specified in the section "Protection diagram and algorithm of operation"; T_1 and T_7 are the signals that the time between the actuations of reed switches 1 and 2 attains Δt_{po} and that 5 ms has expired from the time of actuation of reed switches 5, 13, or 21; t_1, t_2, t_3, t_4, and t_5 are the times counted by timers T_1–T_5; t_{et1} is the time required to detect that the contacts a reed switch have not broken in the same AC half-wave during which they have closed (10 ms); O_1' and O_1'' are the variables with values equal to the results of calculation of the 1st and 7th terms in (10). Note that timer T_1 counts the time between the actuations of reed switches 1 and 2; timers T_2 and T_5, the duration of the closed state of the reed switches; timer T_3, the time for which signals R_1 and T_1 are memorized, and timer T_4, the time after the actuation of reed switches 5, 13, or 21.

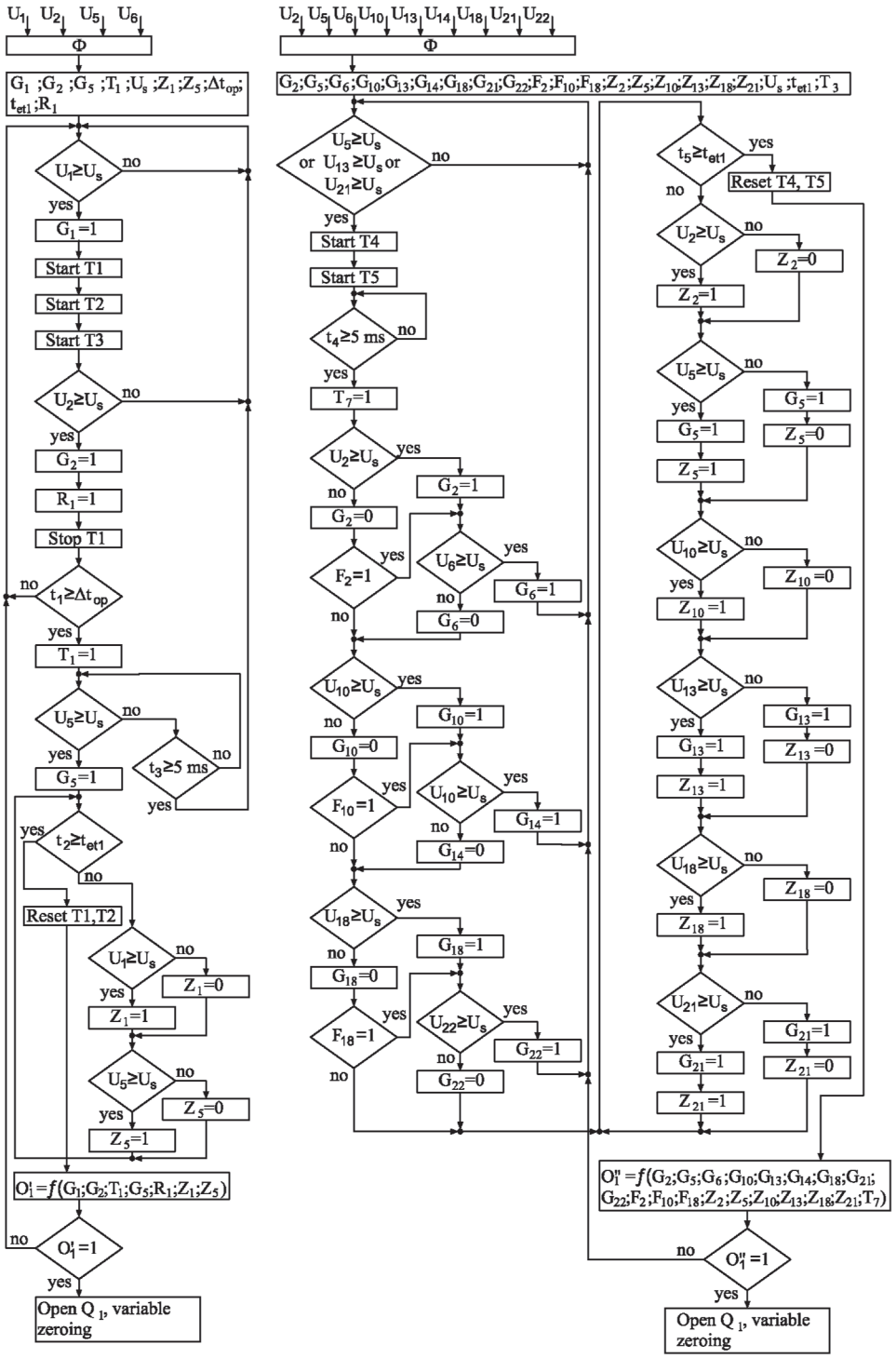


Fig. E.1. Block diagram of algorithm for implementation of the (a) 1st and (b) 7th terms in (10).

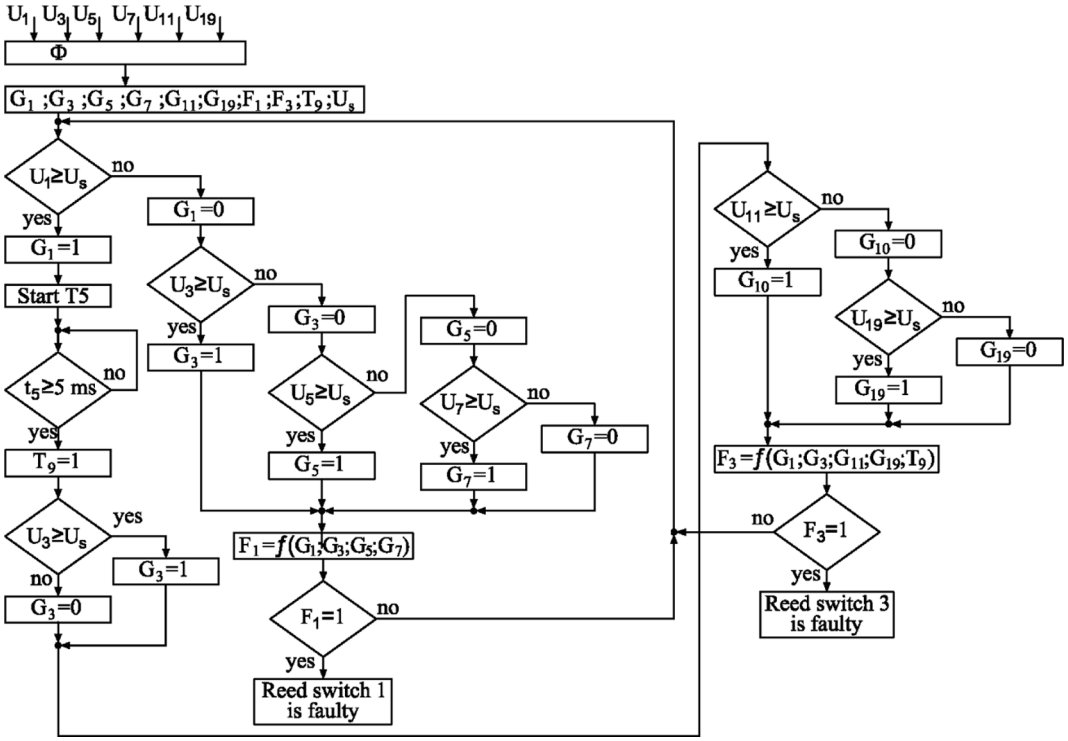


Fig. E.2. Block diagram of algorithm for detecting malfunction of reed switches 1 and 3.

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